

Review Report: PS1 Engineering Drawing Verification

Reviewer: Anirudh Kandwal

Stage 1 – House Footprint Generation

Overview

The proposed architecture uses a two-agent workflow:

- Agent 1 generates Python/Shapely code for generating a house footprint layout.
- Z3 Solver verifies generated geometry against predefined logical constraints.
- Agent 2 analyzes verification failures and provides corrective feedback.
- The loop continues until constraints are satisfied, after which a DXF file is generated.

Strengths

1. Clear separation between generation and verification.
 - Agent 1 focuses on layout generation.
 - Z3 handles constraint verification.
 - Agent 2 handles feedback and correction.
2. Verification-driven workflow.
 - The design does not blindly trust LLM outputs.
 - Formal verification through Z3 improves reliability.
3. Iterative improvement loop.
 - The feedback mechanism allows correction of invalid layouts instead of restarting the process.
4. DXF generation support.
 - Produces CAD-compatible outputs that align with future project requirements.

Observations and Suggestions

- **Constraint Management**

The current example appears to be tied to a specific set of rules and coordinates.

Questions:

- How will constraints be managed when supporting multiple cities or jurisdictions?
- Will there be a centralized constraint repository or rule engine?

- **Geometry Verification Scope**

- Current verification appears focused on coordinate-level validation.
- Future considerations:
 - Overlap detection
 - Accessibility validation
 - Connectivity checks
 - Structural constraints
 - Room adjacency constraints

- **Prompt Dependency**

The workflow relies heavily on prompt-generated code.

Questions:

- As system complexity increases, should more logic move into deterministic software components?
- Would structured constraint representations improve maintainability?

- **Intermediate Representation**

- A common geometry representation (JSON/Schema) between agents may simplify future integrations and debugging.
-

Stage 2 – Interior Room Layout Generation

Overview

The second stage extends the workflow to interior room generation including:

- Bedrooms
- Bathrooms
- Corridors
- Living Rooms
- Kitchen Spaces

within the generated building footprint.

Strengths

1. Logical extension of Stage 1.
 - Builds on the verified house footprint.
2. Reuse of verification architecture.
 - Existing generation → verification → feedback loop remains applicable.
3. Modular design.
 - Interior generation is separated from footprint generation.

Observations and Suggestions

- **Scalability**

As room count increases, constraint complexity will grow significantly.

Future considerations:

- Room adjacency constraints
- Minimum room dimensions
- Door accessibility
- Corridor connectivity
- Natural lighting constraints
- Ventilation requirements

- **Multi-floor Support**

- The current workflow appears focused on a single-floor layout.
- Future considerations:
 - Multi-floor planning
 - Stair placement

- Bathroom stacking
 - Structural alignment across floors
- **Permit Package Alignment**

Based on the broader project vision, future stages may require support for:

- Elevation views
- Roof design
- Mechanical room planning
- Crawl space generation
- Square footage calculations
- Site plan generation

Overall Assessment

The solution demonstrates a strong foundation for an agentic generation-verification workflow. The use of Z3-based validation and iterative correction is a positive architectural decision.

The next major challenge will be scaling the approach from footprint and room generation toward a complete permit-generation pipeline, including multi-floor layouts, elevation design, mechanical planning, and jurisdiction-specific compliance requirements.